Analytics Platform Deployment Audit

Work log: everything completed before the Spark investigation

Scope

This report consolidates the deployment review, documentation reading, server verification, operational checks, and template/dbt contract analysis completed before the focused Apache Spark connector investigation began.

Environment reviewed

The existing analytics platform on host **cp2** (172.31.102.188), and the separate template repository on the user's Mac. This is an audit record, not a fresh deployment or a claim that the platform was redeployed.

Safety and change boundaries

- No account passwords were changed. Correct values were assigned to shell environment variables only when needed for read-only verification.
- No changes were pushed to GitHub.
- No deployment playbooks were rerun against the existing server.
- The requested reboot check was skipped; no reboot was performed.
- No destructive data operations, restores, or restore drills were performed.

Prepared from the recorded session notes and command outputs available in the conversation. Values are point-in-time observations and may change.

1. Executive summary

The existing deployment was already running when reviewed. The work focused on understanding the intended architecture and prerequisites, checking the documented deployment state, exercising a synthetic Airflow-to-ClickHouse run, inspecting backups and monitoring, and reviewing the template repository's transformation contract and dbt implementation before starting the Spark-specific research.

Overall: the checks performed support that the existing stack was functioning for the tested paths. They do not prove every user-facing workflow, OIDC browser login, backup restore, or the future Spark replacement.

2. Task scope and working rules

- Employer's task: clone both analytics-platform repositories, run the existing implementation as-is, document issues, read docs, and develop a separate Spark transformation pipeline from the template without replacing the default implementation prematurely.
- Eyob's assigned slot: Apache Spark instead of dbt, using analytics-platform-template.
- The template's Spark README explicitly says the implementation is not implemented and has not been run.
- Changes must remain local until employer review; no push to GitHub.
- No password changes, no Kubernetes adoption, and no unnecessary server changes.
- The platform on cp2 is shared with another teammate; findings should be reported rather than “fixed” by changing host identity or server setup.

3. Access, repositories, and GitHub finding

Environment access

- Connected to target host cp2 (172.31.102.188) through the jump server at 196.189.50.42.
- Target account was admin; working directory was /home/admin.
- Recorded OS: Ubuntu 24.04.5 LTS; Python 3.12.3; Git 2.43.0.
- The target's home directory already contained platform-related environments and files, including analytics-platform-venv, bsis-venv, mlt-venv, .docker, .kube, and OpenBao-related token/key files. Sensitive values are intentionally omitted from this report.

Repositories

- Single-server repo: betsegaw-projects/analytics-platform-single-server.
- Template repo: betsegaw-projects/analytics-platform-template.
- The single-server repository was cloned to ~/analytics-platform-single-server on the server after local GitHub CLI authentication on the Mac allowed an HTTPS clone into /tmp.

Access issue

From cp2, SSH-based GitHub cloning failed with “Permission denied (publickey)” and `ssh -T git@github.com` also failed. HTTPS access to github.com worked. The server had an SSH key pair, but the recorded checks did not establish that its public key was authorized for the GitHub organization/repository. This is an access/configuration finding, not a reason to expose or copy credentials.

4. Architecture and deployment documentation reviewed

The single-server README describes this data path:

```
Source → Meltano in Airflow → landing bucket (SeaweedFS) → raw ClickHouse → dbt → modelled ClickHouse
→ Superset
```

Documentation read: README; docs/01-prepare-the-server.md; docs/02-deploy.md; docs/03-check-the-deployment.md; docs/04-operate.md; docs/05-backup-and-restore.md; docs/06-troubleshoot.md; docs/decisions.md; docs/glossary.md; docs/what-was-left-out.md; plus template slot-swap and transformation contract documentation.

Important documented prerequisite

- The preparation guide calls for Ubuntu 24.04, at least 4 CPU / 16 GB RAM, and at least 100 GB free on the filesystem used for data_root.
- It warns not to reuse a server already running PostgreSQL, ClickHouse, Docker, or Airflow.
- The deployment guide describes a multi-step Ansible deployment and changes to services, Docker, files, kernel limits, firewall, and SSH settings. Those playbooks were not rerun during this audit because the host already had a live deployment.

5. Initial server-state observations

Disk-space interpretation: the observed 51 GB free was below the guide's 100 GB free-space requirement. The handoff note had previously recorded 57 GB free; the audit records the actual command observation available during this session.

6. Deployment verification: checks 0–11

The following summarizes the documented checks that were actually run. Check 12 (reboot) was explicitly skipped.

Check 0 was a read-only secret retrieval and shell setup. A wrong shell variable assignment was corrected by reassigning the environment variable; no account password was changed. The OpenLineage warning about a missing namespace appeared during the successful DAG run and did not block completion.

For check 10, `systemctl is-active pgbackrest` returned inactive, while `pgbackrest info` reported the `main` stanza as OK with backups. The service-unit state should be recorded separately from the successful backup inventory; no restore test was performed.

7. Operational and troubleshooting review

Operations checks

- Keycloak health was UP.
- Airflow DAG import errors were empty.
- Superset health was OK.
- Prometheus targets were up; Alertmanager was OK; Grafana database was OK.
- Caddy was active and served HTTP 200 on localhost.
- Backup timers were enabled and waiting.
- ClickHouse backup remote showed two backups.
- No `.snap` OpenBao snapshot was found in the specific OpenBao data directory and home-directory locations checked. This is a verification gap only; it does not by itself prove OpenBao is broken.

Troubleshooting checks

- `host.docker.internal` resolved from the expected context.
- The relevant DAG was unpaused.
- Airflow PATH was usable.
- The facility DAG contained `append_env=True`.
- The dbt mount was read-only.
- Cosmos execution used tag-based selection for the Ethiopian period dimension.
- ClickHouse restore documentation used `--tables`.

Observed documentation discrepancy

The unsigned SeaweedFS S3 endpoint returned HTTP 400 during the audit, while the operations documentation was understood to expect HTTP 403. Record this as a behavior/documentation discrepancy for follow-up; no configuration was changed to force a particular response.

Not tested

- Browser-based OIDC/SSO end-to-end login.
- Actual backup restore or disaster-recovery drill.
- Reboot survival check.
- Every dashboard/query or external integration beyond the listed smoke checks.

8. Template repository and transformation contract

The template repository was reviewed locally on the Mac. The Spark slot README says the Spark implementation is not implemented and nothing in that slot has been run. The intended experiment is to replace only the transformation slot, keep the default dbt implementation unchanged, and verify the same transformation contract against the real server before considering it complete.

Transformation contract

- Landing input: `s3://landing/facility_reports/run_id=/data.jsonl`.
- Inputs include raw tables/seeds, orchestration run ID, and connection settings.
- `raw_facility_reports`: append-only; order `(report_id, updated_at)`; all records per run retained; `source_run_id` recorded; duplicates intentionally retained.
- `stg_facility_reports`: typed and renamed view; tenant normalized to lower case/trimmed.
- `fct_facility_reports`: `ReplacingMergeTree(updated_at)`; current row per `report_id` when queried with `FINAL`.
- `mart_facility_reports_monthly`: total value grouped by tenant, org unit, data element, and month.
- `dim_period_ethiopian`: 9,496 rows.
- Tests must exit 0 when all pass; the day-5 negative fixture must fail `assert_facility_reports_value_non_negative`.
- The current DAG sequence is extract → land → dbt_load_raw → dbt_transform → dbt_test → reconcile.
- The period-dimension DAG is separate and uses Cosmos selection `tag:dim_period_ethiopian`.
- Secrets are retrieved from OpenBao; state is not intended to live outside the warehouse.
- Contract proof after five landing objects: raw 355 rows, fact current rows 120, mart 40 rows, total 24,402; day 5 produces one negative-value test failure; reconciliation 120/120.

dbt implementation inspected

- Raw model reads landing JSONEachRow from ClickHouse `s3()` using landing credentials and an explicit schema; it includes nullable `data_quality_flag` and `source_run_id`.
- Staging model normalizes tenant values.
- Fact model uses `ReplacingMergeTree(updated_at)` with `order_by(report_id)`; current-row behavior is evaluated with `FINAL`.
- Monthly mart uses `fct_facility_reports FINAL`, joins Ethiopian periods, groups and sums.
- Custom test looks for negative values in `fct_facility_reports FINAL`; the day-5 fixture contains `-999`.
- The Ethiopian period CSV had 9,497 lines including its header, corresponding to 9,496 data rows.

9. Airflow implementation inspected before Spark research

- Airflow image: `apache/airflow:3.3.1`.

- Installed stack includes astronomer-cosmos 1.15.1, dbt-core 1.11.14, dbt-clickhouse 1.10.2, AWS/PostgreSQL/FAB providers, boto3, and clickhouse-connect.
- Meltano 4.2 is installed in a separate virtual environment.
- No Java, Spark, or PySpark was installed in the Airflow image/source configuration reviewed.
- Compose includes Airflow API server, scheduler, DAG processor, and triggerer with LocalExecutor; no Spark service is defined.
- The dbt directory is mounted read-only at `/opt/airflow/dbt`.
- The current facility DAG uses BashOperators/dbt because the runtime run ID varies; Cosmos is used for the Ethiopian period dimension.
- The template slot-swap guide says to copy `docs/experiments/TEMPLATE.md` to a dated slot/tool experiment file, document the question and motivation, create the slot implementation, meet the contract, syntax-check configuration, run only the slot and dependencies, then validate the contract on the real server. Keep the stub marked not implemented until real-server validation succeeds.

10. Findings, risks, and follow-up register

11. Boundary: where the Spark work began

This report intentionally stops before the detailed Apache Spark connector investigation. The next workstream began by reviewing the Spark slot README and the template contract, then inspecting the ClickHouse Spark connector v0.9.0 source for Spark 3.5 compatibility, ReplacingMergeTree support, batch writes, and truncate behavior. The separate connector research and its unresolved write-mode question belong in a follow-up Spark investigation report, not in this pre-Spark work log.

At the point the Spark investigation began, no Spark implementation had been added to the template, no Spark pipeline had been run, and no GitHub push had been made.

12. Suggested concise handoff

“I audited the existing cp2 deployment and read the single-server and template documentation before starting the Spark slot investigation. The documented checks 0–11 passed for the paths tested, including a successful `verify-1` Airflow run, warehouse checks, Superset, monitoring, Caddy, and backup inventory. Check 12/reboot, browser OIDC, and restore drills were not performed. Findings to report include 51 GB free versus the documented 100 GB prerequisite, GitHub SSH authentication failure from cp2, SeaweedFS unsigned S3 HTTP 400 vs. expected 403, no OpenBao snapshot found in the locations checked, pgBackRest service inactive despite healthy backup inventory, the init-container count mismatch, and the cp1/cp2 handoff mismatch. No passwords were changed, no deployment was rerun, and nothing was pushed to GitHub.”